Detection of sparsity in neural responses

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1 Motivation

Very few investigators record visual texture responses from IT due to the lack of a strong, coherent response. *A priori*, we should expect a sparse response in IT. There are a few ways that sparseness may appear in IT:

1. There are a few IT neurons whose responses to the texture families tend to be strong and reliable.
2. For each texture family, there are a few neurons whose response is reliable, but the responsive neurons for each family may differ.
3. For some texture families, there are a few neurons whose response is reliable.
4. Same as (3), but for each specific image (sample ID).
5. Same as (3), but for each specific image.

Because reliable sites are identified by correlation to the same stimuli, it is possible to consistently miss patterns of sparsity outside of (1). In this new analysis, I want to visualize the reliability of neurons in a way that is not dependent on the previous criteria for reliable site. Furthermore, I want to compare this approach across all regions recorded. Since scenario (5) is most general, we want to focus on approaches to characterize the neural response that would work in this situation.

2 Plan

The most basic analysis is to perform a two sample t-test for each neuron's response for a specific image compared to the response to blank images. Since there are 6 repetitions, this should work well enough. Furthermore, it would be useful to visualize each component of the t-test. For example, if in IT, we may see that the difference in means for samples in a family tend to be consistently positive but the differences are not significant due to higher variance. This may indicate that there is some response at the family level and that we could detect a significant difference if the trials were pooled at the family level rather than the specific image level. Regardless, it would be worthwhile to do this analysis both at the specific image level and the texture family level.

3 Data

See `20260625.qmd` for reference for how data were processed. Note that the window of averaged responses depends on the criteria for reliability. For now, I will proceed with the already calculated average, but note this potential source of bias, especially for IT which does not have many reliable sites.

Unlike the previous notebooks, analysis will be *not* be limited to reliable sites. For now, we will still only consider images with naturalistic structure compared to blank images.

4 Methods

4.1 Two-sample t-test

To compare the mean firing between blank and non-blank images, Welch's t-test will be used. The statistic t is found as

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{s^2_{\overline{X}_1} + s^2_{\overline{X}_2}}}, \quad s_{\overline{X}_i} = \frac{s_i}{\sqrt{n_i}}$$

where $\overline{X}_i$ is the sample mean, and $s_{\overline{X}_i}$ is the standard error, s_i is the sample standard deviation, and n_i is the sample size. The t-distribution's degrees of freedom can be approximated as

$$\nu \approx \frac{\left(s^2_{\overline{X}_1} + s^2_{\overline{X}_2}\right)^2}{\frac{s^4_{\overline{X}_1}}{n_1 - 1} + \frac{s^4_{\overline{X}_2}}{n_2 - 1}}.$$

For simplicity, this will be implemented by `t.test()`.

5 Results

5.1 Blank image baseline response

The average response and the standard deviation of response to blank images will be pooled over all the repetitions of blank images, of which there are 600 for each session. The blank images are labeled by ID (1 to 100) and repetitions (1 to 6), though I believe this is arbitrary. Density plots of the mean blank responses and the standard error of blank responses for each site and region are displayed in Figure 1. V1 responses tend to exhibit lower standard deviations and lower mean firing rate, but the differences between V2, V4, and IT are less noticeable. V2 seems to have more variable mean firing in response to blank images.

5.2 Qualitative per-site response to images

5.2.1 Distribution when pooled over non-blank images

The per-site mean and standard deviations for firings for all non-blank images (pooled) can be seen in the left of Figure 2. The means are reported as the difference with the blank images. The spread of

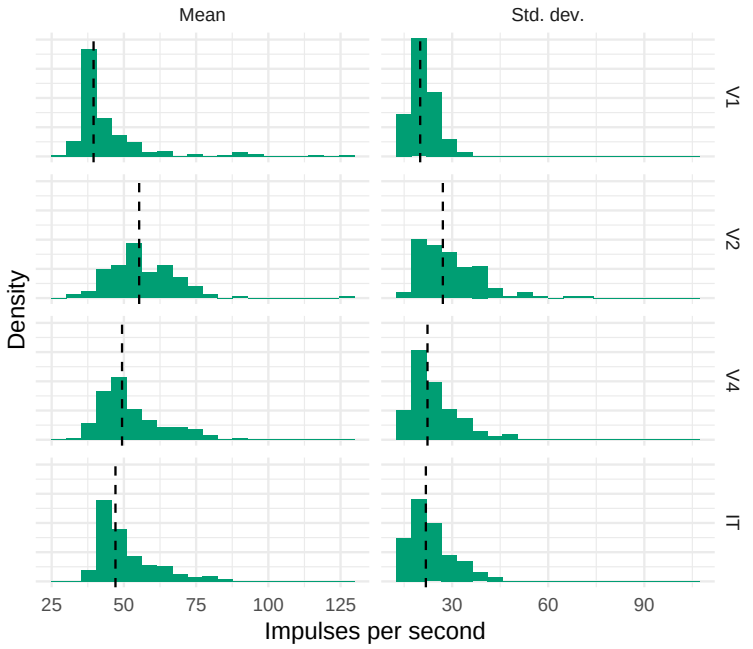


Figure 1: Histogram of the mean response to blank images and SE of response to blank images for each region. The number of observations differs for each region, so the y-axis has been replaced with density rather than count. Dashed lines indicate the median. V1 shows lower mean firing rates and standard deviations.

the mean difference for IT is very narrow with a lower median than the other regions. The high mean difference for V2 is consistent with previous work. V1 and V4 are comparable, which is somewhat surprising considering previous work by Lee et al. showing that coding for naturalistic texture families is stronger in V4. These results could still be consistent if the noisy textures produce much smaller average responses in V4, though we are not necessarily examining this currently.

5.2.2 Distribution when pooled over specific image repetitions

Figure 2 (right) contains the difference in means and the standard deviations for the specific images, so pooling is only done over repetitions. With more data points, the distributions tend to look less irregular overall, though the trends in the medians are similar to when pooling is performed over the sites in **?@fig-sites**. The distribution of the difference in means for IT appears shifted to the left and more centered around zero.

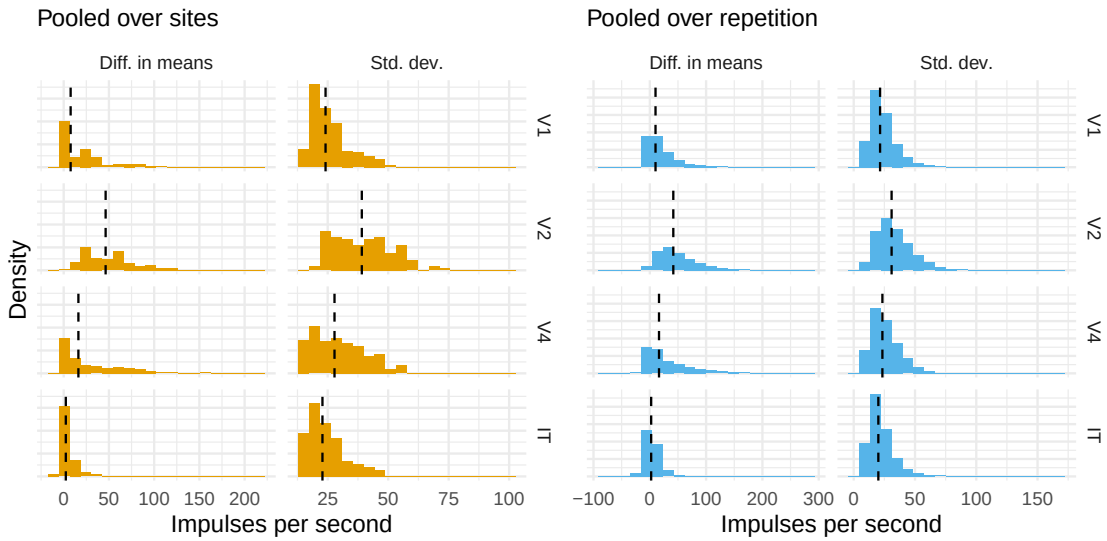


Figure 2: (Left) Densities of the mean response (difference with blank images) and standard deviations of responses to pooled non-blank images for each region. Dashed lines indicate the median. IT shows lower and less variable differences in mean firing rate and lower standard deviations. (Right) Results with pooling only over repetition. The overall trends are similar to the left plot, though the IT differences in means seem more centered around zero.

5.2.3 Firing in specific combinations of site and image

In Figure 3, the heat map of mean differences and standard deviations helps visualize the per-site and per-image distribution of firing patterns. Even with this basic visualization, the divisions between each array tend to be clear, with the exception of V2. For instance, in V4, the second array (i.e., the 96 columns in the center of the heat maps) shows markedly lower differences in means and standard errors.

As a heuristic, vertical striping in the heat maps indicates per-site differences and horizontal striping in the heat maps indicates per-image differences. V2 and V4 show more between-site and between-image variation, consistent with previous results on decoding texture families in these regions.

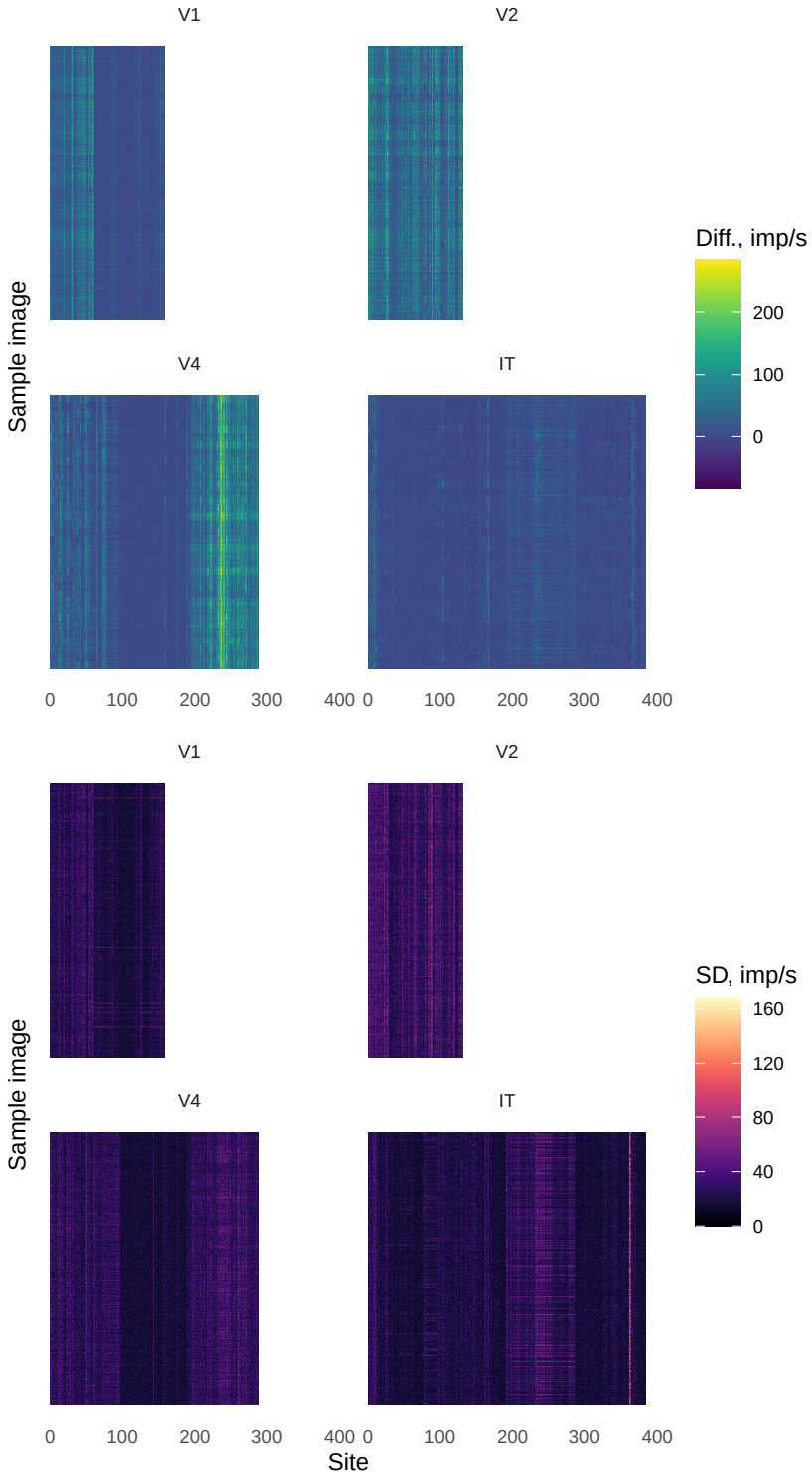


Figure 3: (Top) Heat map of the differences in means between sample images and the blank images for each site. (Bottom) Standard deviations in firing over repetitions of each sample image for each site. V2 and V4 show more between-site and between-image variation in firing patterns.

5.3 Two sample t-test results

The heat map of t–statistics (top) and p-values from the t-tests in Figure 4 displays similar information

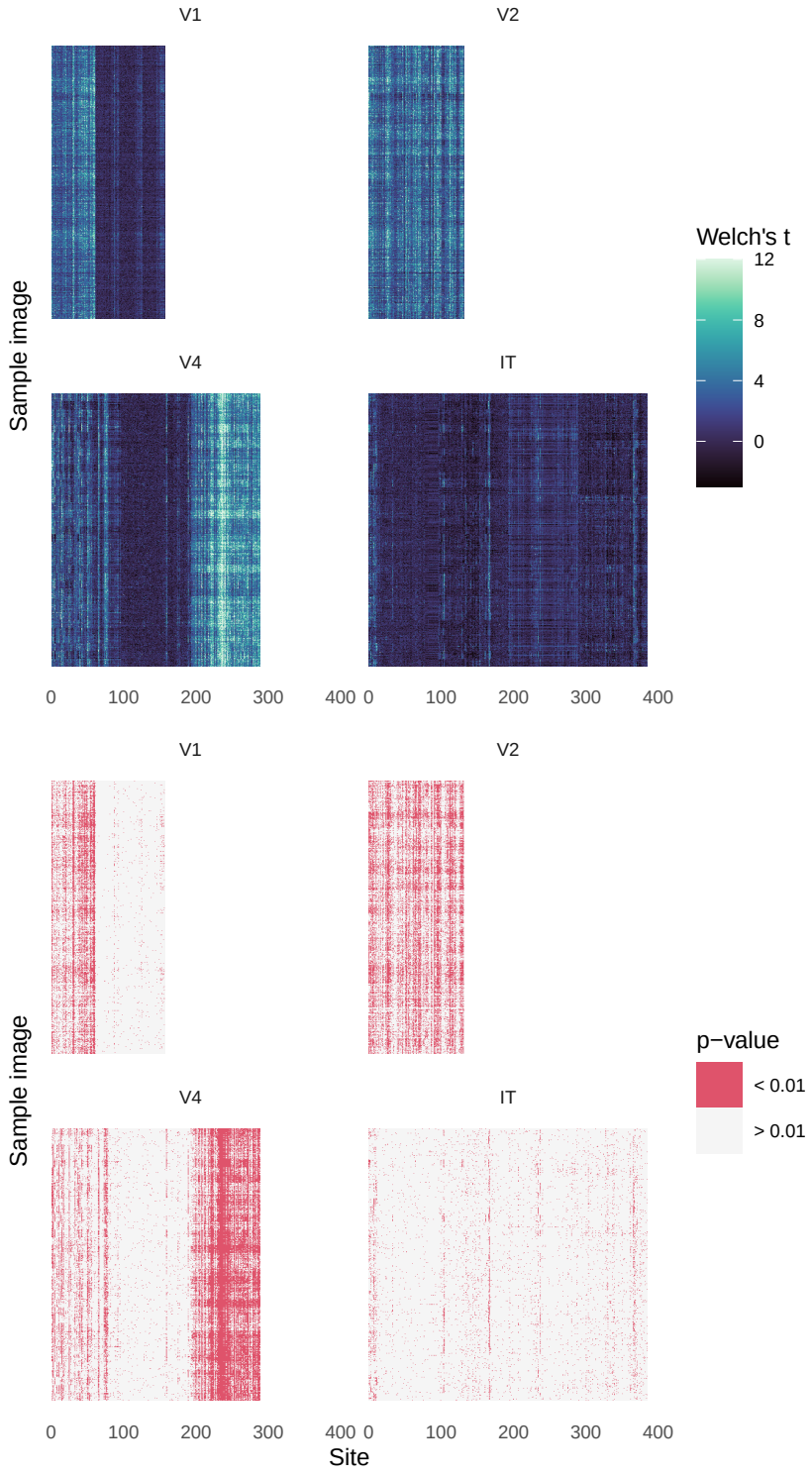


Figure 4: (Top) Heat map of the t-statistics calculated for each combination of site and image. The degrees of freedom are identical fir eacg test. The scale has been truncated to (-3, 12). (Bottom) Corresponding p-values, simplified to indicate whether values are below or above a cutoff of 0.01. The graphs are similar to the difference in means plots, though the p-value plot more clearly shows the vertical, per-site firing consistency in V4 and the sparseness in IT.